The simulator already knew: what the AIs inside a materials-discovery loop actually know

Team: [add names] · OSI Hackathon 2026 · code: github.com/bruce249/osi-discovery-agent · video: [video link]

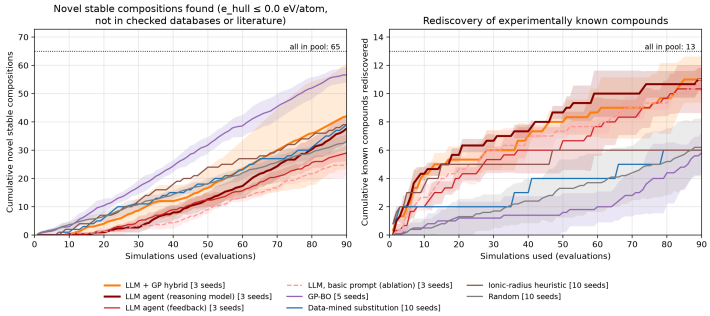
Problem

Li3MX6 / Li2MX6 halides (Li3YCl6, Li3InCl6, Li2ZrCl6) are leading solid electrolytes for all-solid-state batteries. Metal, halide and ordering choices make a large space and each candidate costs a simulation, so **deciding what to simulate** is the bottleneck. We built 236 candidates (123 compositions) by substituting Materials Project parents: 22 trivalent and 5 tetravalent metals, Cl/Br, ordered mixed-halide and mixed-metal variants.

Method

Loop: a strategy proposes 10 candidates, each is simulated, results feed back, until the budget is spent; known/novel labels stay hidden. Every evaluation is a CHGNet relaxation scored by its energy above a convex hull of 513 MP reference phases relaxed with the same potential (validation: MAE 14 meV/atom vs DFT over 399 compounds; all DFT-stable ternaries within 0.03 eV/atom). Hit = e_hull ≤ 0.0 eV/atom (on or below the hull). The LLM agents (gpt-4.1-mini; reasoning arm gpt-5-mini) see all unevaluated candidates and the full (id, description, e_hull) history and return 10 picks with a chemical reason each. Baselines: random, ICSD substitution probabilities (Hautier et al.), ionic-radius similarity, GP-BO on Magpie features; all with the same budget (90 of 236).

Result



With 90 simulations from a pool of 236 candidates (65 novel stable compositions hidden in it), GP-BO found the most novel stable compositions, 56.6 ± 2.8 versus 33.1 ± 2.8 for random: random search needs 1.7x as many simulations to match it (1.5x on all stable compositions), and the data-mined substitution baseline reaches 1.2x.

strategy (@90 sims)	seeds	stable found	novel found	known rediscovered	speed-up vs random (stable / novel)
GP on MACE latent	5	69.2 ± 5.7	55.8 ± 6.0	9.0 ± 2.1	1.6x / 1.7x
GP-BO	5	65.8 ± 4.6	56.6 ± 2.8	5.8 ± 1.5	1.5x / 1.7x
GP on scrambled-MACE latent	5	60.0 ± 6.4	48.6 ± 5.2	6.8 ± 1.2	1.4x / 1.5x
LLM + GP hybrid	3	58.0 ± 16.5	42.0 ± 18.5	11.0 ± 1.6	1.3x / 1.3x
LLM agent (reasoning model)	3	54.3 ± 3.9	37.7 ± 4.7	11.0 ± 0.8	1.3x / 1.1x
LLM agent	3	43.7 ± 1.7	29.0 ± 2.8	10.3 ± 1.7	1.0x / 0.9x
Data-mined substitution	10	48.0 ± 0.0	39.0 ± 0.0	6.0 ± 0.0	1.1x / 1.2x
Random	10	43.2 ± 2.9	33.1 ± 2.8	6.2 ± 2.0	1.0x / 1.0x

Looking inside both AIs: the simulator already knew, the LLM remembers

The LLM's edge is memory. On real reaction-yield screens (our team's benchmark, github.com/bruce249/osihackiitd; Buchwald-Hartwig, Ahneman et al. 2018, 5 substrates) the LLM agent reached 0.84 of the best possible yield after 10 experiments with chemical names shown, 0.74 with names hidden and 0.74 for random. Here, LLM agents alone rediscover the most known compounds but are only 0.8-1.1x random on new ones: in both domains the advantage is prior knowledge, not search.

Reading the simulator's mind. With the probing protocol of our MACE interpretability study (linear probes, scrambled-weight control, splits grouped by chemistry), MACE-MP-0 features of the *unrelaxed* candidates predict their post-relaxation stability on unseen chemistries: rank correlation 0.85, AUC 0.92, R^2 0.67, vs 0.80 / 0.90 / 0.41 for the same network with scrambled weights and 0.72 / 0.84 for Magpie composition features (shuffled labels: AUC 0.49). A GP searching in this latent space found 69.2 stable compositions, the most of any arm, and reached half of the novel ones 1.85x faster than random; the scrambled control found 60.0. The signal peaks after the first message-passing layer, training adds it mainly on Li and halide sites, textbook descriptors explain only 42% of it, and erasing 10 stability directions (INLP) drops the probe AUC to 0.40 and the GP to 61.2 stable (random erasure: 64.4; within seed spread). The same network also learned iron's redox state from energies alone (companion probe, github.com/Intick/osi-draft: Fe moments on unseen chemical systems R^2 0.75 vs 0.19 geometry, 0.06 scrambled).

Dynamic stability. Finite-displacement phonons (CHGNet + phonopy; tiers from our Lattice studio) for the top candidates: 2/15 free of imaginary modes, 12 with soft modes between -1 and -0.1 THz (framework tilts: Li carries only 0%-24% of them while being 30% of the atoms; they need larger supercells or DFT) and 1 unstable (Li2ZrBr6, -2.8 THz, flagged).

New candidates (CHGNet + MACE-MP-0 cross-check; 15/15 agreed within 0.05 eV/atom)

formula	prototype	CHGNet e_hull	MACE e_hull	agreed	databases	literature
Li2CeBr6	Li2ZrCl6-type	-0.133	-0.032	yes	absent*	none found
Li3YbBr6	Li3InCl6-type	-0.070	-0.003	yes	absent*	none found
Li3Ho0.5Yb0.5Cl6	Li3InCl6-type	-0.060	-0.037	yes	absent*	none found

Validation and limitations

ML potentials approximate DFT (CHGNet reference MAE 14 meV/atom; MACE 21 meV/atom); stability margins here are a few meV/atom, comparable to that error, so hits are predictions, not proof. Yb compounds come out most stable with CHGNet but much less so with MACE (Yb reference energies are known to be delicate in MP-trained models); Li2CeX6 needs Ce4+, which is not guaranteed to survive. Ordered approximants of partly disordered phases; 0 K thermodynamics only (no ionic conductivity or electrochemical window); campaigns replay precomputed evaluations. The brief's 0.03 eV/atom hit threshold makes 92% of the pool 'hits', so we used e_hull <= 0 (0.03/0.05 in summary.json). *'Novel' = not found in MP, GNoME, JARVIS, OQMD (partly reachable) and literature searches; the Alexandria server was unreachable during the run and is reported as unchecked.